Joseph Brazina

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EDUCATION

Virginia Polytechnic Institute and State University | **Blacksburg, VA**
M.Eng. in Industrial & Systems Engineering (Human Factors Engineering & Ergonomics) | *Expected May 2028*
Relevant Coursework: Human Factors System Design, Human Factors of Visual Display Systems
James Madison University | **Harrisonburg, VA**
B.S. in Psychology, Minor in Sociology | *Graduated May 2026*
GPA: 3.50 | Major GPA: 3.75

HUMAN FACTORS & UX PROJECTS

Configural Conflict Display for Air Traffic Control | **Visual Display Systems Course** | *Fall 2026*
• Designed an animated en-route radar display that merges 5 flight variables into one emergent conflict shape
• Built 7 Figma frames from real OpenSky ADS-B data with reusable components and Smart Animate transitions
• Iterated through 4 versions using class critique, adding redundant color and size cues for color-blind users

Nintendo Switch 2 Home Screen Redesign | **Human Factors System Design Course** | *Fall 2026*
• Designed the wireframe and lo- and hi-fi interactive Figma prototypes on a 3-person usability evaluation team
• Translated usability-test pain points into design changes, each justified with human factors principles
• Tuned text, icon, and focus-state contrast across 6 home-screen themes to keep controls legible

Mobile App Usability Evaluation | **Self-Directed Project** | *Fall 2026*
• Mapped a mobile app's task flow with hierarchical task analysis and found 3 issues via heuristic evaluation
• Prioritized redesign recommendations by linking each issue to high-risk task steps
• Translated each recommendation into a Figma mockup; planning a moderated usability test to validate findings

Multiple Resource Theory Dual-Task Experiment | **Experiment Build** | *Fall 2026*
• Built a browser-based experiment testing whether Tetris suffers more alongside visual vs. auditory tasks
• Programmed 5 counterbalanced conditions logging performance, response times, and NASA-TLX workload

Traffic Signal Detection Experiment | **Human Information Processing Course** | *Fall 2026*
• Designed 80 go/brake stimuli varying glance time (200-516.7 ms) and deutan color-deficiency severity (24-100%)
• Analyzed accuracy, false-alarm rates, and d′ in JASP to evaluate signal visibility for color-deficient drivers

RESEARCH EXPERIENCE

Research Assistant | **Human IMPaC-T Lab, Virginia Tech** | *Oct 2026 - Present*
• Support a study on how air- vs. bone-conduction 40 Hz audio affects attention, collecting participant data

Principal Investigator | **Human Development Lab, James Madison University** | *Jan 2025 - May 2026*
• Designed and managed a 200+ participant IRB-approved study on sleep, mental health, and cognition
• Wrote and deployed 3 Qualtrics surveys, then analyzed results in SPSS using correlation and regression
• Presented findings to faculty and student audiences at the 2025 and 2026 JMU Psychology Symposium

Research Assistant | **Auditory Perception Lab, James Madison University** | *Aug 2024 - Dec 2024*
• Designed a study on how music affects working memory and ran 20+ computer-based participant sessions
• Configured 43 DirectRT experimental-trial files to implement task conditions and stimulus manipulations

PROFESSIONAL EXPERIENCE

Behavior Technician | **ABA Today, Haymarket, VA** | *May 2025 - Aug 2025*
• Observed client responses in real time and adapted task demands for a caseload of 10+ clients
• Tracked 25+ client learning goals per session and reported progress weekly to parents and BCBAs

SKILLS

HF/UX & Design: Figma (wireframing, lo-/hi-fi prototyping, components, Smart Animate), usability testing, heuristic evaluation, hierarchical task analysis, display design, data visualization, accessibility, Tableau
Research & Analysis: experimental design, human-subjects research, signal detection theory, NASA-TLX, Qualtrics, SPSS, JASP, Excel, t-tests, ANOVA, regression, mediation analysis, DirectRT

HONORS & PROFESSIONAL ACTIVITIES

Human Factors & Ergonomics Society | Institute of Industrial & Systems Engineers | Psi Chi | Cum Laude Honors